

CHE 201 HW 1
Due: 11.59pm on 09/06/2019

1. (10 points)

For problems 1.1–1.10, match the appropriate definition in the right column with each term in the left column.

- 1.1.  ___ Boundary
- 1.2.  ___ Closed system
- 1.3.  ___ Control volume
- 1.4.  ___ Extensive property
- 1.5.  ___ Intensive property
- 1.6.  ___ Process
- 1.7.  ___ Property
- 1.8.  ___ State
- 1.9.  ___ Surroundings
- 1.10.  ___ System

- A.** The condition of a system as described by its properties
- B.** A region of space through which mass may flow
- C.** Whatever is studied
- D.** A transformation from one state to another
- E.** A property whose value for an overall system is the sum of its values for the parts into which the system is divided
- F.** Everything external to the system
- G.** A fixed quantity of matter
- H.** A property whose value is independent of the size or extent of a system and may vary from place to place within the system at any moment
- I.** Distinguishes the system from its surroundings
- J.** A macroscopic characteristic of a system to which a numerical value can be assigned at a given time without knowledge of the previous behavior of the system

2. (8 points)

WP Perform the following unit conversions:

- a.** 1 L to in.³
- b.** 650 J to Btu
- c.** 0.135 kW to ft · lbf/s
- d.** 378 g/s to lb/min
- e.** 304 kPa to lbf/in.²
- f.** 55 m³/h to ft³/s
- g.** 50 km/h to ft/s
- h.** 8896 N to ton (=2000 lbf)

3. (5 points)

1.19 WP A closed vessel having a volume of 1 liter holds 1.2×10^{22} molecules of nitrogen gas. For the nitrogen, determine (a) the amount present, in kmol and kg, and (b) the specific volume, in m³/kmol and m³/kg.

4. (5 points)

1.20 WP The specific volume of 5 kg of water vapor at 1.5 MPa, 440°C is 0.2160 m³/kg. Determine (a) the volume, in m³, occupied by the water vapor, (b) the amount of water vapor present, in gram moles, and (c) the number of molecules.

5. (10 points)

1.24 WP Figure P1.24 shows a gas contained in a vertical piston–cylinder assembly. A vertical shaft whose cross-sectional area is 0.8 cm^2 is attached to the top of the piston. Determine the magnitude, F , of the force acting on the shaft, in N, required if the gas pressure is 3 bar. The masses of the piston and attached shaft are 24.5 kg and 0.5 kg, respectively. The piston diameter is 10 cm. The local atmospheric pressure is 1 bar. The piston moves smoothly in the cylinder and $g = 9.81 \text{ m/s}^2$.

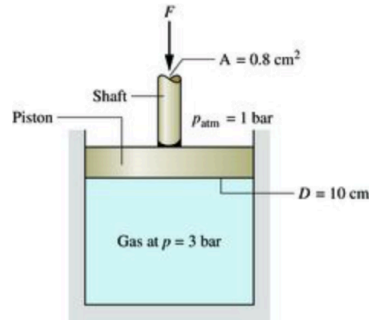


FIGURE P1.24

6. (8 points)

1.25 A gas contained within a piston–cylinder assembly undergoes four processes in series:

Process 1–2: Constant-pressure expansion at 1 bar from $V_1 = 0.5 \text{ m}^3$ to $V_2 = 2 \text{ m}^3$

Process 2–3: Constant volume to 2 bar

Process 3–4: Constant-pressure compression to 1 bar

Process 4–1: Compression with $pV^{-1} = \text{constant}$

Sketch the processes in series on a p – V diagram labeled with pressure and volume values at each numbered state.

7. (8 points)

1.26 WP Referring to Fig. 1.7,

- if the pressure in the tank is 1.5 bar and atmospheric pressure is 1 bar, determine L , in m, for water with a density of 997 kg/m^3 as the manometer liquid. Let $g = 9.81 \text{ m/s}^2$.
- determine L , in cm, if the manometer liquid is mercury with a density of 13.59 g/cm^3 and the gas pressure is 1.3 bar. A barometer indicates the local atmospheric pressure is 750 mmHg. Let $g = 9.81 \text{ m/s}^2$.

8. (10 points)

1.28 WP As shown in Figure P1.28, the exit of a gas compressor empties into a receiver tank, maintaining the tank contents at a pressure of 200 kPa. If the local atmospheric pressure is 1 bar, what is the reading of the Bourdon gage mounted on the tank wall in kPa? Is this a *vacuum* pressure or a *gage* pressure? Explain.

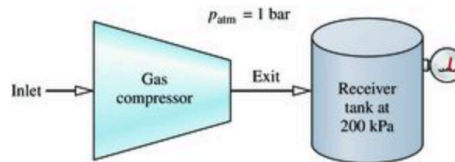


FIGURE P1.28

9. (8 points)

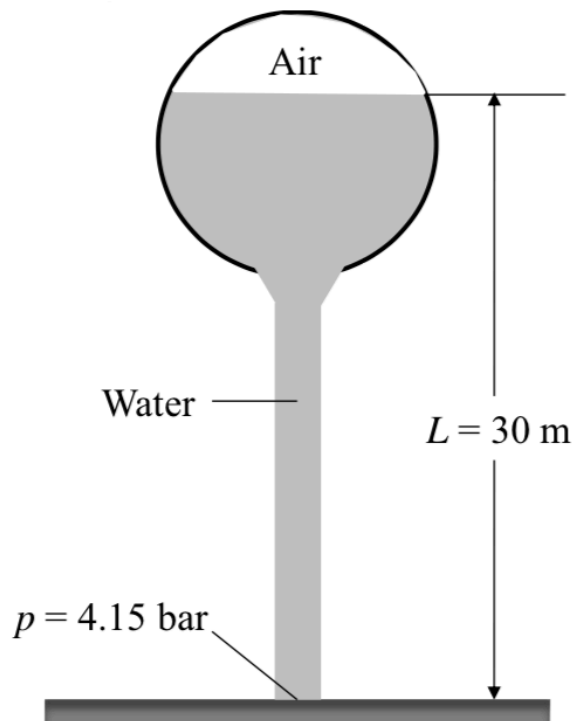
1.32 Show that a standard atmospheric pressure of 760 mmHg is equivalent to 101.3 kPa. The density of mercury is $13,590 \text{ kg/m}^3$ and $g = 9.81 \text{ m/s}^2$.

10. (8 points)

1.33 WP A gas enters a compressor that provides a pressure ratio (exit pressure to inlet pressure) equal to 8. If a gage indicates the gas pressure at the inlet is 5.5 psig, what is the absolute pressure, in psia, of the gas at the exit? Atmospheric pressure is 14.5 lbf/in.^2 .

11. (8 points)

1.37 WP If the water pressure at the base of the water tower shown in Fig. P1.37 is 4.15 bar, determine the pressure of the air trapped above the water level, in bar. The density of the water is 10^3 kg/m^3 and $g = 9.81 \text{ m/s}^2$.



12. (12 points)

1.1D In the United States today, nearly all of our electricity is produced by fossil-fuel power plants burning coal or natural gas, nuclear power plants, and hydroelectric power plants. Using the Internet, determine the percent contributions of these types of electricity generation to the U.S. total. For each of the four types, determine at least three significant environmental considerations associated with it and how such environmental aspects affect the respective plant design, operation, and cost. Write a report with at least three references.